P3: Generating Land Use Land Cover Map using Supervised Classification Techniques

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Contents

Chapter 1

Introduction

This project shall be a study on the creation of a LULC (Land Use Land Cover) map of the Mumbai Metropolitian Region. We shall be analysing the spatial features of Mumbai in the time frame of 1 January 2025 till 1 May 2025 and classifying them into their constituent classes. The software used for the same is Google Earth Engine.

1.1 Objective

To generate a land use and land cover (LULC) map using supervised classification methods by processing satellite imagery with the help of QGIS and Google Earth Engine.

1.2 Remote Sensing

We obtain the data for our study via Satellite Imagery, which is a subcategory of Remote Sensing.

Remove Sensing refers to the non-physical acquisition of information about an object or a phenomenon. Some examples include SONAR, Satellite Imagery, and RADAR. It forms a fundamental aspect of our daily life, aiding in Weather Forecasting, GPS navigation, Soil Health Assessment, etc.

In Satellite Imaging, we record the transmitted and reflected light from the object on the sensor. Different sensors are built for different resolutions and EM bands. Due to the inherent property of the object, it may selectively choose to reflect certain frequencies of light (this property being known as its spectral signature) which can be useful in identifying it.

1.3 Geographic Information System (GIS)

The data collected from Remote Sensing is usually processed in a Geographic Information System (GIS).

Geographic Information System is the combination of Computer Hardware, Software, Personnel, Data and its processing methods which is used to manipulate, analyse and present information tied to a spatial location. It provides the location, condition, trends & patterns and modelling associated with a spatial location. Some examples of GIS Software include QGIS and Google Earth Engine.

1.4 Land Use and Land Cover Classification

LULC Classification refers to the process of categorizing Earth's surface into various classes e.g. Vegetation, Water Bodies, Urban Areas, etc. It consists of two aspects:

- Land Use - Human utilization of land
- Land Cover - Physical Attributes of land

LULC Classification has important applications:

- Urban Planning
- Environmental & Natural Resource Management
- Disaster Response

1.5 Organization of The Report

This chapter provided you with a basic introduction to the topics discussed in this project. In the upcoming chapters, we shall see the detailed explanation of the methodology, data, and results.

In Chapter 2, we take a look into the Study Area of this project i.e. Mumbai, its previous relevant statistics and our defined area. In Chapter 3, we discuss the data collected from two Satellites used for the analysis - Sentinel 2A & Landsat 8. In Chapter 4, we look into an overviwe of Supervised Classification. In Chapter 5, we delve into the methodology used and some aspect of Machine Learning. In Chapter 5 & 6, we finally look into the Results and Conclusion.

We have also created a Github Repository for the code and data.

Chapter 2

Study Area

Our study area is a part of the Mumbai Metropolitan Region (MMR). It is one of the most populous regions of the world. The prominent city of MMR is Mumbai, which is the financial capital of India and the state capital of Maharashtra. It is also the most populous city of the nation.

2.1 About Region

The Salsette Island is located just off the Konkan coast of Maharastra, housing the cities of Mumbai, Thane and Mira-Bhayander. It is bounded by the Vasai Creek, Thane Creek, Ulhas river and the Arabian Sea. The Mumbai Metropolitan Region consists of Mumbai, its satellite towns on the Salsette island, and some towns on the mainland Konkan coast.

The current city of Mumbai is formed by the systematic reclamation of land around a heptesia (group of seven islands). Hence, it is now on a narrow peninsula on the southwest of the Salsette Island (located off the Konkan coast), surrounded by the Arabian Sea, Thane Creek and Vasai Creek. Many parts of the southern part of the city lie just above sea level, while the northern part is hilly with dense vegetation.

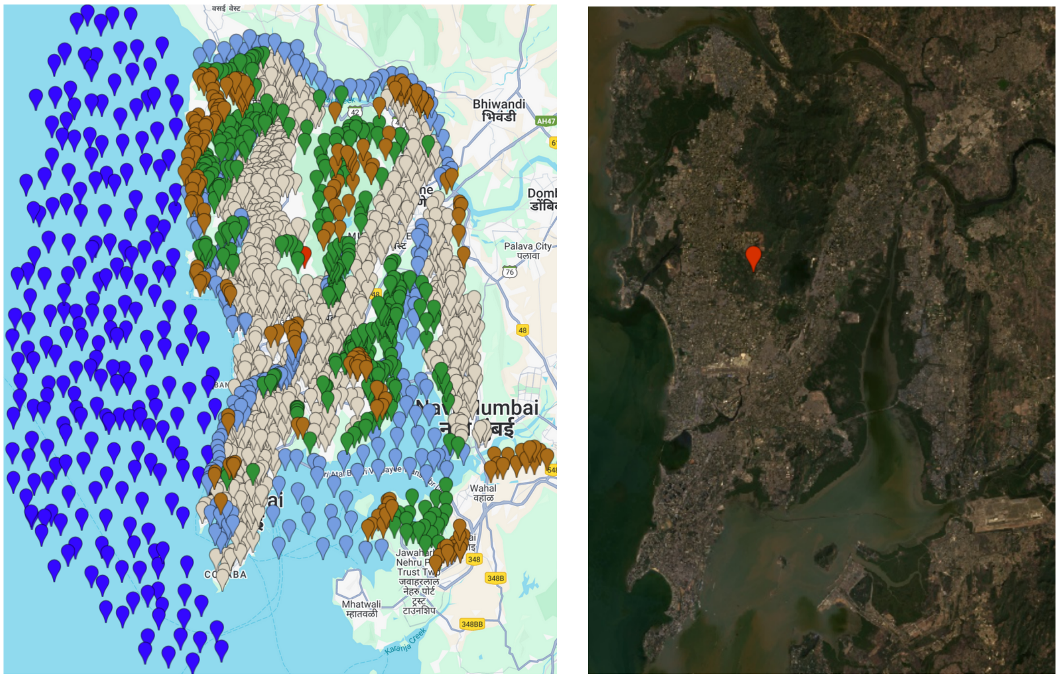
The study area (2.1(b)) chosen extends to the complete Salsette Island and a partial inclusion of Navi Mumbai.

2.2 Area of Interest

We decided to divide the AOI into 5 classes:

1. Oceans
2. Urban Areas
3. Vegetation
4. Water Bodies
5. Other Lands

We created a bunch of points (2.1(a)) across the map of island demarking as different classes. The statistics for the same is as 2.1.



(a) Mumbai AOI (b) Mumbai Region

Fig. 2.1 Mumbai Region and the defined AOI

Ocean	O	232
Urban Area	U	854
Water Bodies	W	235
Vegetation	V	314
Other Land	O	201

Table 2.1 Features Distribution

Chapter 3

Data Used

For recording the images reflected from the surface, we use the satellite images which are corrected for maximizing the *Surface Reflectance*. We also ensure that the masking by the clouds are minimum by ensuring the images are taken during non-Monsoon time (Jan - May 2025) and additionally correcting for any residuals. Finally we apply any required scaling to the raw images.

We primarly used the data of two satellites - Sentinel 2A and Landsat 8 due to their prominent and extensive dataset and orbital period.

3.1 Sentinel 2A

It is a sun-synchronous satellite operated by the **European Space Agency (ESA)** as part of the Copernicus Programme. It has high temporal (10 days) and spatial resolution (10 m) with a swath width of 290 km. It has a multi-spectral sensor which can measure the bands mentioned in 3.1(a). For this project, we utilise the bands B2, B3, B4, B5, B6, B7, B8, B11, & B12 for the study.

3.2 Landsat 8

It is a sun-synchronous satellite operated by **NASA** and the **U.S. Geological Survey (USGS)** as part of the Landsat Programme. It has a temporal resolution of 16 days, a spatial resolution of 30 m and a swath width of 185km. Its sensor can measure the bands mentioned in 3.1(b). It additionally has a Thermal Infrared sensor as well. For this project, we utilise the bands B2, B3, B4, B5, B6, & B7 for the study.

3.3 Remote Sensing Indices

Although it suffices to use the True Color RGB images for our purpose, we decided to augment the data by including the SWIR and NIR (Near Infrared) bands as well. These bands are used in calculating the SAVI, NDBI, NDVI & NDWI indices which can be used to further segregate the classes. The indices are explained below.

3.3.1 Soil-Adjusted Vegetation Index (SAVI)

Soil Adjusted Vegetation Index (SAVI) is used to correct Normalized Difference Vegetation Index (NDVI) for the influence of soil brightness in areas where vegetative cover is low.

Formula -

$$SAVI = \frac{(NIR - Red)(1 + L)}{(NIR + Red + L)} \tag{3.1}$$

3.3.2 Normalised Difference Built-up Index (NDBI)

The Normalized Difference Built-up Index (NDBI) uses the NIR and SWIR bands to emphasize manufactured built-up areas

Formula -

$$NDBI = \frac{(SWIR - NIR)}{(SWIR + NIR)} \tag{3.2}$$

3.3.3 Normalised Difference Vegetation Index (NDVI)

The normalized difference vegetation index is a widely used metric for quantifying the health and density of vegetation using sensor data.

Formula -

$$NDVI = \frac{(NIR - Red)}{(NIR + Red)} \tag{3.3}$$

3.3.4 Normalised Difference Water Index (NDWI)

It is used to monitor changes related to water content in water bodies, using green and NIR wavelengths.

Formula

$$NDWI = \frac{(Green - NIR)}{(Green + NIR)} \tag{3.4}$$

Band	Resolution	Central Wavelength	Description
B1	60 m	443 nm	Ultra Blue (Coastal and Aerosol)
B2	10 m	490 nm	Blue
B3	10 m	560 nm	Green
B4	10 m	665 nm	Red
B5	20 m	705 nm	Visible and Near Infrared (VNIR)
B6	20 m	740 nm	Visible and Near Infrared (VNIR)
B7	20 m	783 nm	Visible and Near Infrared (VNIR)
B8	10 m	842 nm	Visible and Near Infrared (VNIR)
B8a	20 m	865 nm	Visible and Near Infrared (VNIR)
B9	60 m	940 nm	Short Wave Infrared (SWIR)
B10	60 m	1375 nm	Short Wave Infrared (SWIR)
B11	20 m	1610 nm	Short Wave Infrared (SWIR)
B12	20 m	2190 nm	Short Wave Infrared (SWIR)

(a) Sentinel Bands

Band Number	µm	Resolution
1	0.433–0.453	30 m
2	0.450–0.515	30 m
3	0.525–0.600	30 m
4	0.630–0.680	30 m
5	0.845–0.885	30 m
6	1.560–1.660	30 m
7	2.100–2.300	30 m
8	0.500–0.680	15 m
9	1.360–1.390	30 m
10	10.6-11.2	100 m
11	11.5-12.5	100 m

(b) Landsat Bands

Fig. 3.1 Bands per Satellite

Chapter 4

Supervised Classification

Supervised Classification is a Machine Learning paradigm which refers to the process of an algorithm learning to classify data points according to the labels defined by the user. Some examples of the algorithms include Random Forest, SVM, CART, etc.

The algorithms that we shall see in this project are CART and Random Forest.

The key components in Supervised Classification are:

1. Features: These are the input variables used for prediction (e.g., the different spectral bands and indices from a satellite image).

2. Labels (or Classes): These are the correct output categories that you are trying to predict (e.g., "Urban," "Vegetation," "Water").

3. Training Data: A subset of your data used to train the model.

4. Test/Validation Data: A separate subset of labeled data that is held back and used to evaluate the model's performance and accuracy after it has been trained.

A model is trained by feeding it a large amount of training data to help it identify the patterns in the data. Once it has been trained, it can predict the correct label for new unseen data.

4.1 Classification and Regression Tree (CART)

A Classification and Regression Tree, or CART, is a type of decision tree algorithm that can be used for both classification (predicting a category) and regression (predicting a continuous value).

It creates a tree based on the data, with each branch consisting of a subset of the data. The algorithm finds the best possible feature and value to split the data into branches until we reach the leaves with final predictions.

Strength: CART models are very easy to interpret and visualize. You can follow the path of decisions from the root to the leaf to understand exactly how the model arrived at a prediction.

Weakness: A single decision tree is often unstable and prone to overfitting. This means it might learn the training data perfectly but perform poorly on new, unseen data because it has created overly specific rules that don't generalize well.

4.2 Random Forest

A Random Forest is an ensemble learning method that dramatically improves the performance of a single decision tree. Instead of relying on one tree, a Random Forest builds a large number of individual decision trees and combines their outputs by taking a majority vote.

Each tree in Random Forest is trained on a random subset of the data. Even within a tree, the split based on features is also done randomly.

Strength: By averaging the predictions of many diverse trees, Random Forests are highly accurate, robust, and much less prone to overfitting than a single tree.

Weakness: They are a "black box" model. Because hundreds of trees are involved, it's no longer possible to easily interpret the decision-making process in the same way you can with a single tree.

4.3 Evaluation Metrics

Evaluation metrics are quantitative measures used to assess a machine learning model's performance. Derived from a *confusion matrix*, they include *Overall Accuracy* (the percentage of correct predictions) and the *Kappa Coefficient*, which measures agreement while accounting for chance, providing a more robust performance assessment.

4.3.1 Confusion Matrix

A confusion matrix is a table used to evaluate the performance of a classification model. It provides a detailed breakdown of how the model's predictions compare to the actual, true labels of a test dataset, showing not just the errors but also the types of errors being made.

For a simple binary (two-class) problem, the matrix is a 2x2 table:

1. True Positive (TP): The model correctly predicted the positive class. (e.g., The model said "Urban" and it was actually Urban).

2. True Negative (TN): The model correctly predicted the negative class. (e.g., The model said "Not Urban" and it was actually Not Urban).

3. False Positive (FP): The model incorrectly predicted the positive class (a "Type I Error"). (e.g., The model said "Urban" but it was actually Vegetation).

4. False Negative (FN): The model incorrectly predicted the negative class (a "Type II Error"). (e.g., The model said "Not Urban" but it was actually Urban).

Chapter 5

Methodology

We shall now pour over the complete methodology of this project. There are four main stages of the project - Dataset Creation, Preprocessing, Training, and Validation & Results.

5.1 Dataset Creation

We must first decide on the Area of Interest. This can be done by marking the region by a polygon or a point. We must also decide on the timeframe to work with.

Once that is done, we must visually analyse the area for the possible classes that are present (e.g Vegetation, Urban Builtup, Oceans, etc). We must then make an educated guess on the distribution of points / features required for each class such that the classifier can correctly predict them (by not over or underfitting the data). For example, MMR has a higher Builtup percentage compared to Vegetation or Water Bodies. Thus, the proportion or the size of Builtup class would be higher than other classes. In this project, we have decided on the split of features as given in 2.1.

Once the distribution is done, we must begin to annotate the date for the features with the labels. The selection of features can be done point-wise or polygon-wise; we did it the former way. The feature selection has to be done in order to avoid any biasness in the data while encompassing all types of data available for the class.

5.2 Preprocessing

Next, we move onto image preprocessing. There are three main preprocessing methods that we use here:

5.2.1 Scaling Corrections

We first select all of the bands corresponding to surface reflectance and apply the scaling factor (as provided in the Landsat 8 guide). We do the same with the thermal bands. We then add the scaled bands back to the image.

5.2.2 Addition of bands

We decide on the appropriate indices for our usecase, and calculate them on the fly using the bands of the satellite. We then add these indices as additional bands on the image to display them.

5.3 Training

Once we have our complete dataset finalized, we can proceed with the training of the classifier. We must decide onto the type of classifier we need (which can be picked based on aforementioned methods). Then, we randomise the data points and split them into training data and validation data (Here, we used the $80 : 20$ split). Finally, we set the classifier into training on the training data.

5.3.1 Validation & Results

We must now test our classifier for its accuracy. We use the validation dataset for this to generate a confusion matrix and the overall accuracy. Based on our satisfaction, we may choose to export any required assets from the platform.

5.4 Overall Algorithm

Algorithm 1 LULC Classification

ROI ← MMR
dataset ← satellite.filterDate($1/1/2024, 1/5/2024$).filterBound(ROI).ScaleCorrections()
image ← dataset.first()
ndbi ← image.normalizedDifference($[\text{SWIR}, \text{NIR}]$)
ndvi ← image.normalizedDifference($[\text{SWIR}, \text{NIR}]$)
ndwi ← image.normalizedDifference($[\text{SWIR}, \text{NIR}]$)
savi ← image.normalizedDifference($[\text{SWIR}, \text{NIR}]$)
image ← image.addBands(ndbi, ndvi, ndwi, savi)
trainingData ← dataset$[: 0.8 * \text{dataset}.length]$
validationData ← dataset$[0.8 * \text{dataset}.length :]$
classifier ← RandomForest(100).train(trainingData)
confusionMatrix ← classifier.predict(validationData)

Chapter 6

Results

We performed various experiments based on three different aspects:

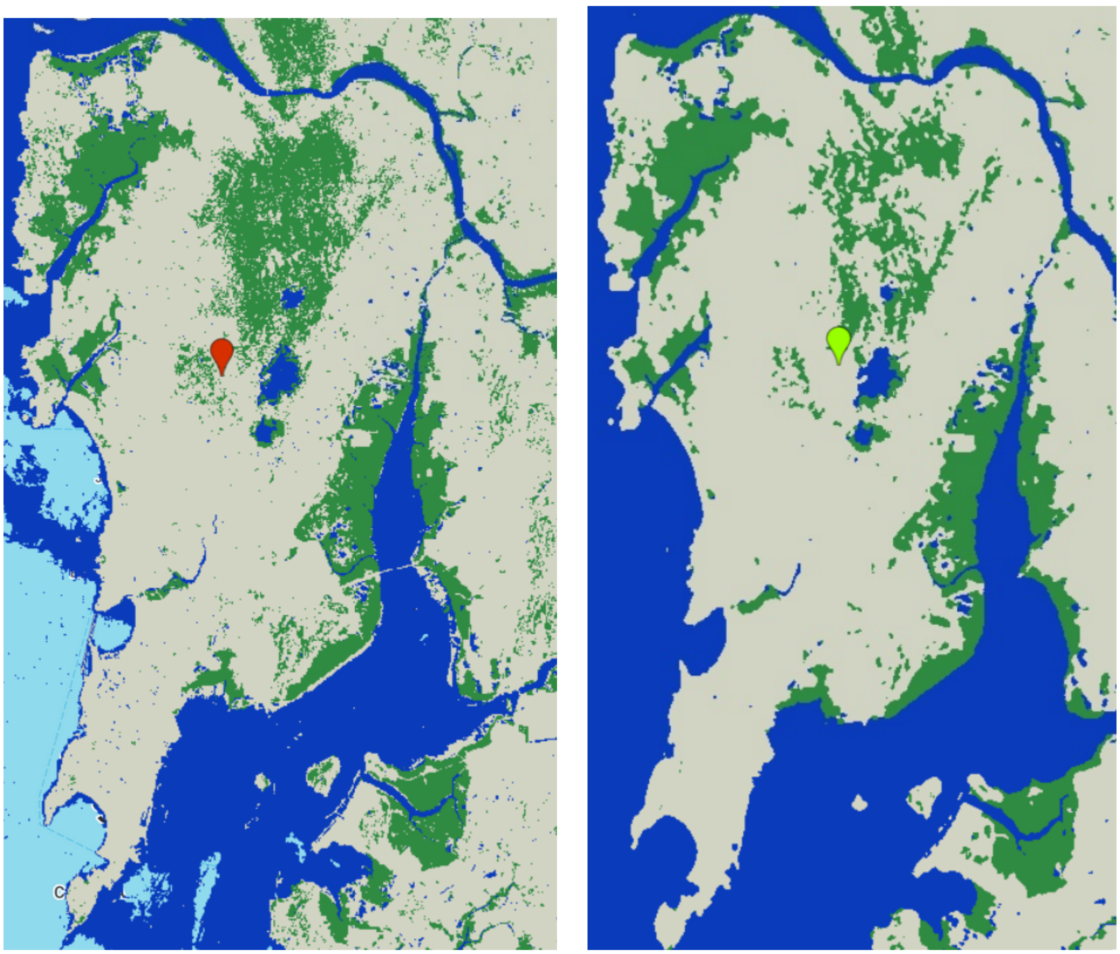
- Satellite: Landsat 8 vs Sentinel 2A
- Classification Method: CART vs RF100
- No of Classes: UVW vs UVWO vs UVWO

We perform a total of 12 experiments for each of the combination and plot the accuracy chart (6.4). We also measured the Per Class Area distribution as given in 6.5. Fig 6.6 is the compiled plot of all confusion matrices for each of the 12 experiments.

We notice the following:

- Random Forest method performs consistently better than CART, as described in the theory.
- We also notice that as we decrease the number of classes to predict, the accuracy also increases. This is due to the increase in inter-class complexities and ambiguities as we increase classes.
- The difference between the two satellites is very minute, and hence can be said to be equally good for the purpose.
- 17% of all land is Vegetation. That includes Forests, parks and overall green cover.
- Nearly a quarter of the land is equivalent to the water bodies in and around the ROI.
- Majority of the land is urban development.

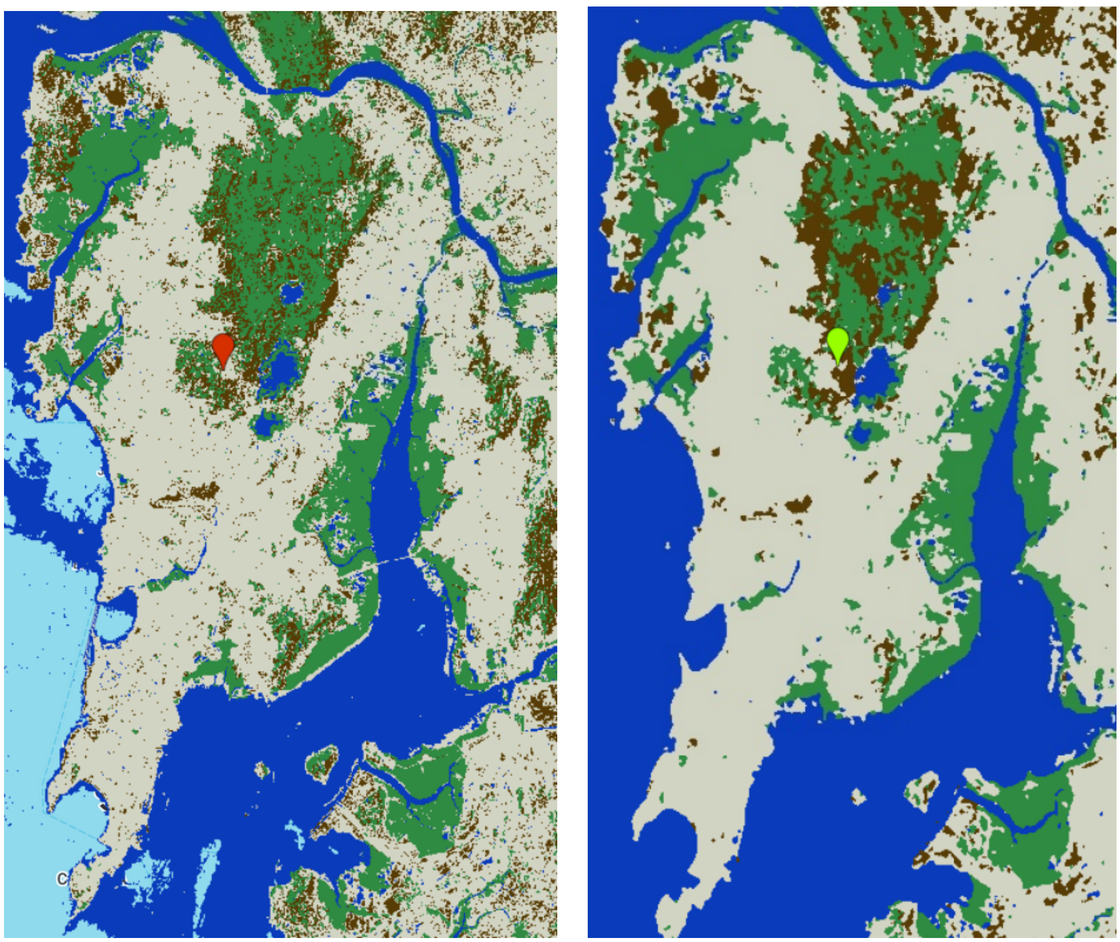
Due to these reasons, we have included six classified images of the AOI (6.1, 6.2 & 6.3).



(a) Classified Image using Landsat 8 & Random Forest for 3 Classes

(b) Classified Image using Sentinel 2A & Random Forest for 3 Classes

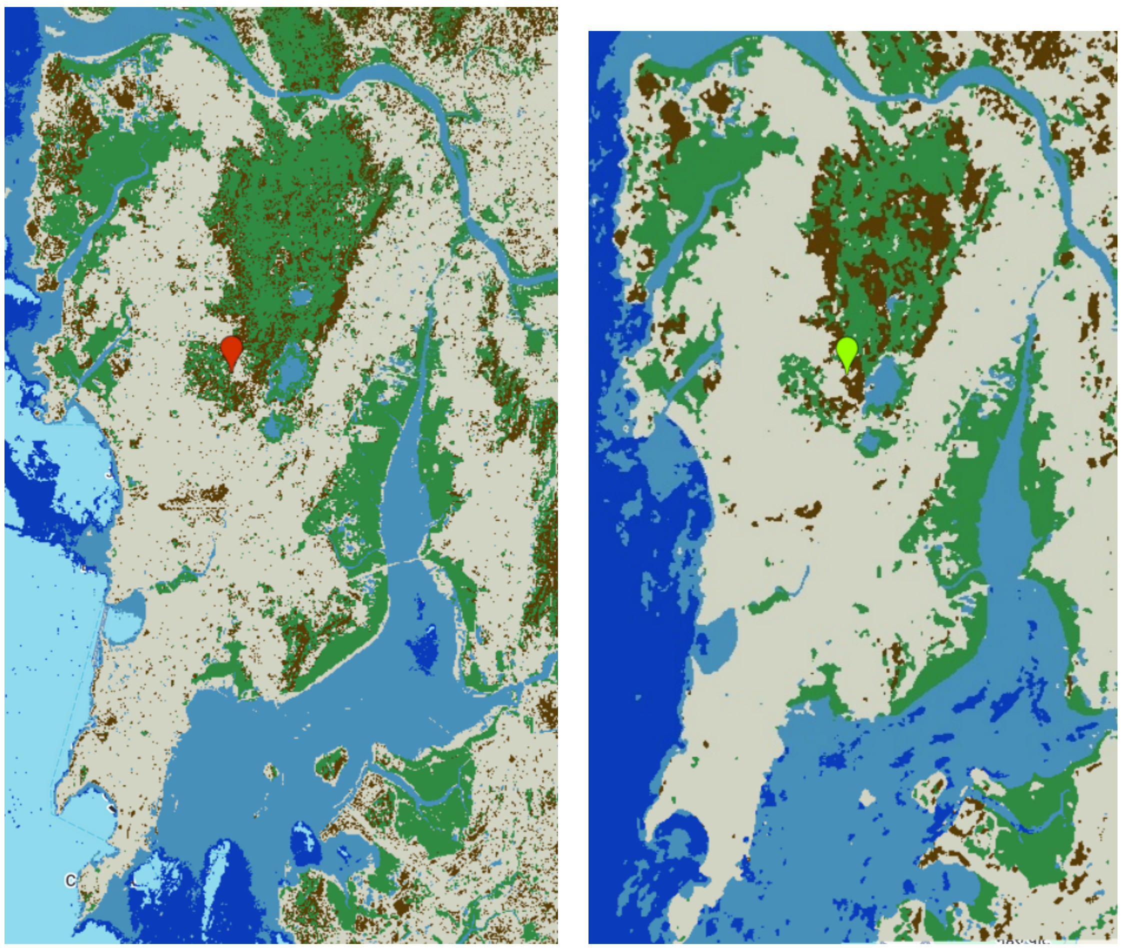
Fig. 6.1 Random Forest Classification for 3 Classes



(a) Classified Image using Landsat 8 & Random Forest for 4 Classes

(b) Classified Image using Sentinel 2A & Random Forest for 4 Classes

Fig. 6.2 Random Forest Classification for 4 Classes



(a) Classified Image using Landsat 8 & Random Forest for 5 Classes

(b) Classified Image using Sentinel 2A & Random Forest for 5 Classes

Fig. 6.3 Random Forest Classification for 5 Classes

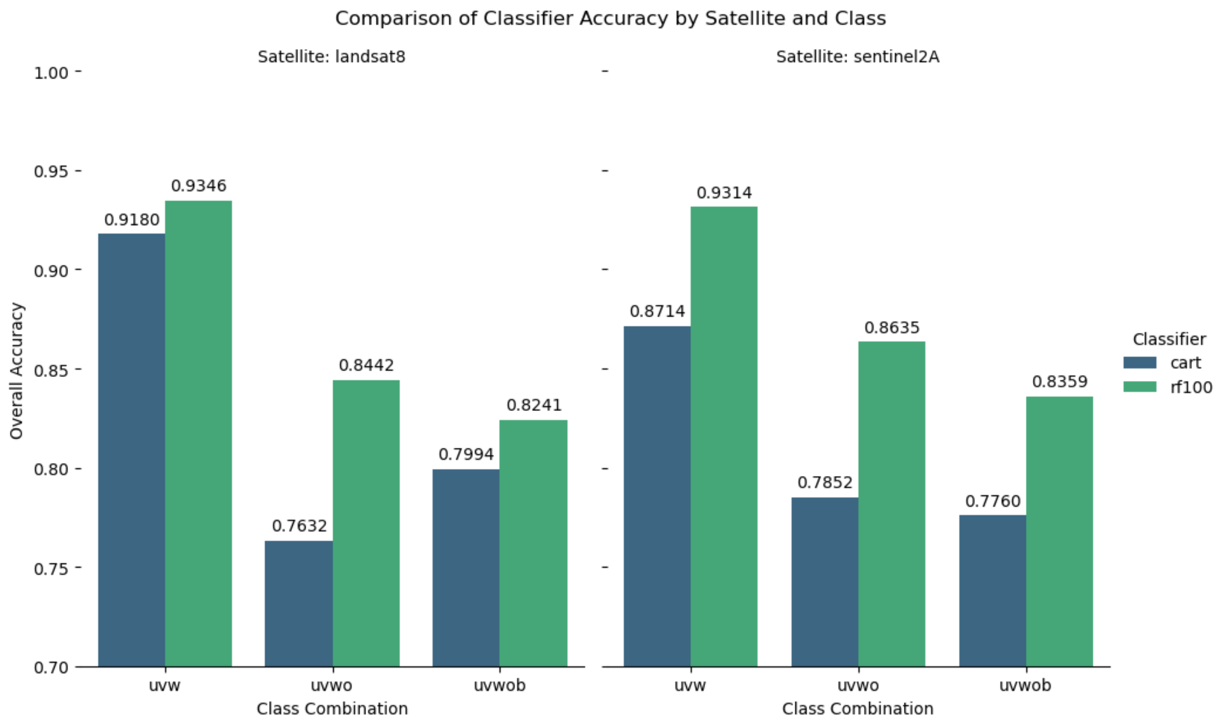
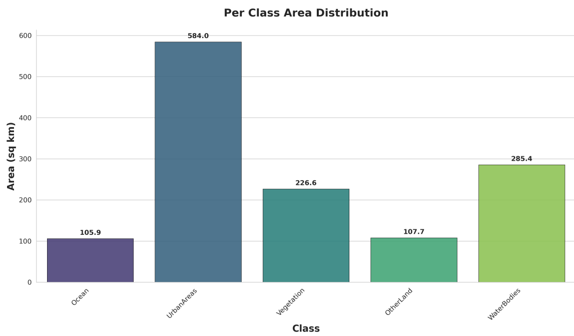
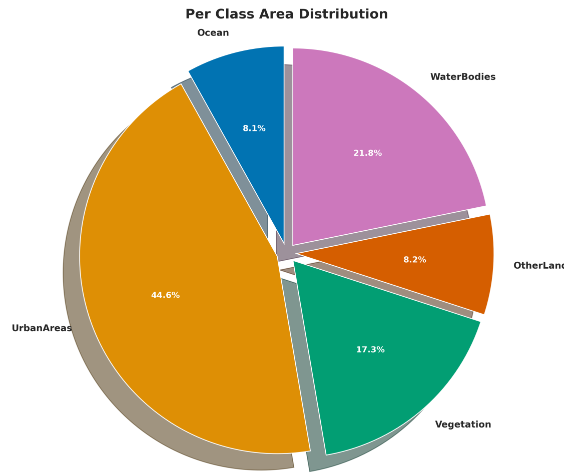


Fig. 6.4 Comparisons

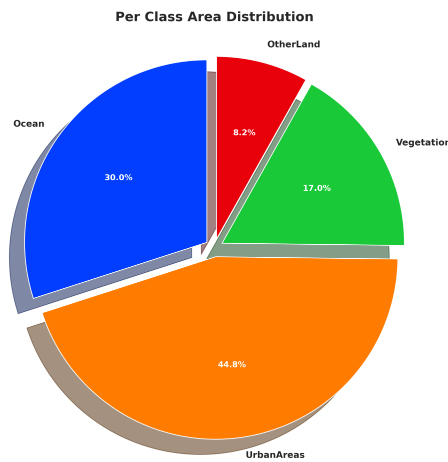


(a) Five Class Area Distribution



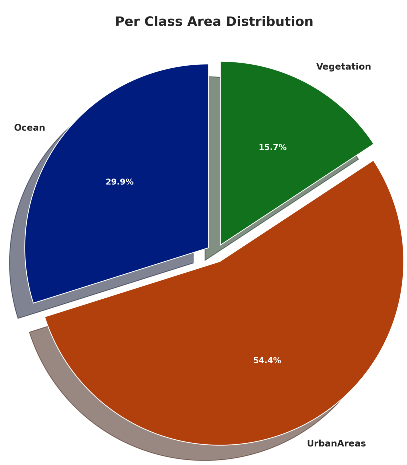
(b) Five Class Area Pie Chart

(c) Four Class Area Distribution



(d) Four Class Area Pie Chart

(e) Three Class Area Distribution



(f) Three Class Area Pie Chart

Fig. 6.5 Area Distribution Per Class

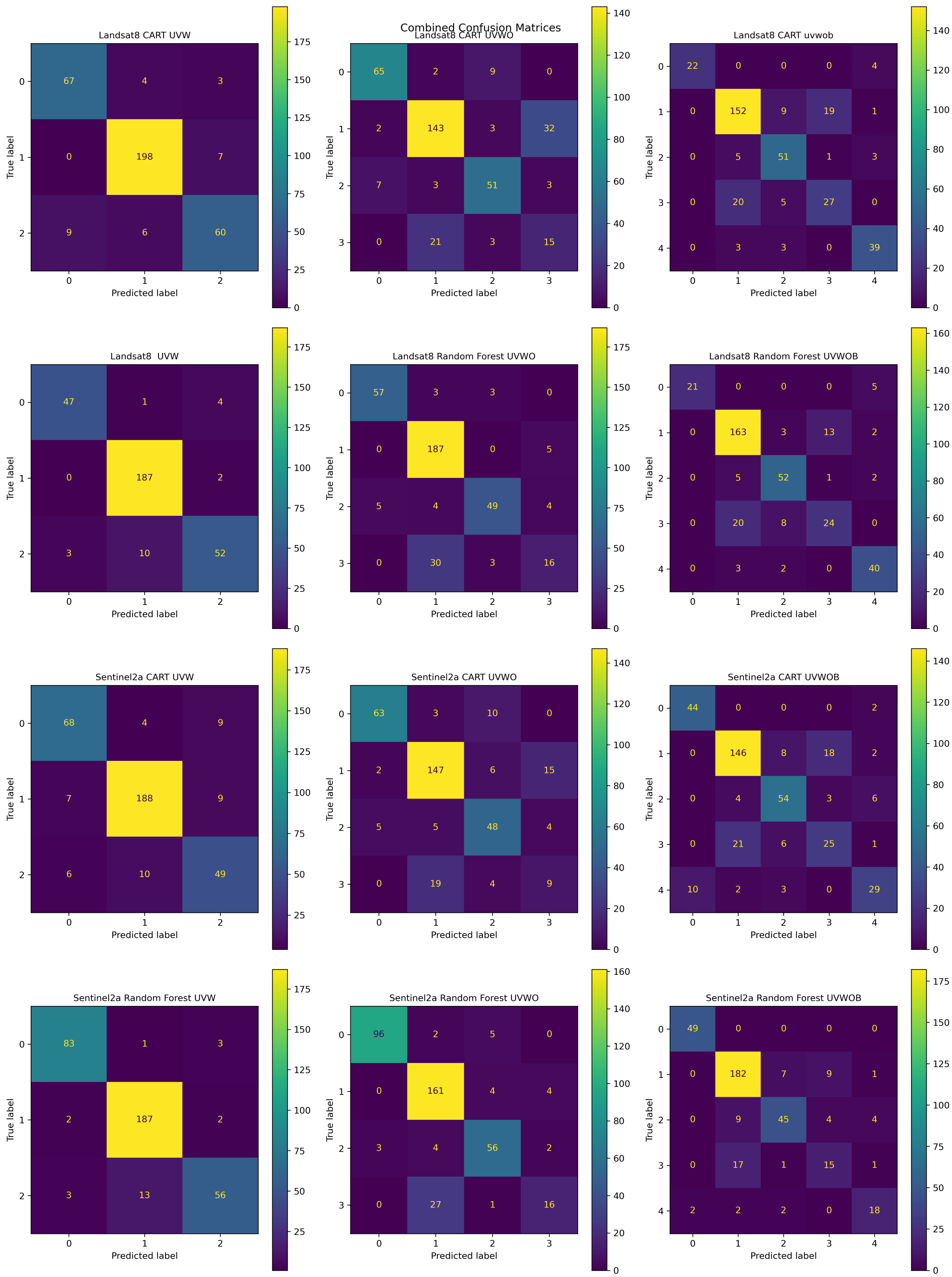


Fig. 6.6 All Confusion Matrices

Chapter 7

Conclusion

In this project we have seen how to perform LULC Classification on a given Area of Interest. We saw how to decide the features and accordingly create our dataset. We then choose an appropriate classifier and train it and get the results. This provides a simple and effective pipeline for generating LULC maps.

Some observations made are:

- Landsat 8 and Sentinel 2A are equally good at LULC data collection.
- There is a considerable overlap of non-urban and urban areas as well as between ocean and inland water bodies classes.
- Nearly a third of the total area is occupied by ocean and the water bodies
- The green cover is 17% of the total area, which falls in the required rate given by the URDPFI.
- The obtained LULC map is important as it can be utilised in Urban Planning and Development and Environmental Modeling.

Some improvements over the current work are:

- Improvement in Resolution: Usage of a better dataset or satellite with a higher resolution can greatly improve the accuracy of the LULC classification.
- Usage of better Classification: The limitation of Google Earth Engine is that we cannot use Deep Nueral Networks for classification. A further issue is they require a huge amount of data to be trained on. Although there is a solution for this, this project focused on classical Machine Learning techniques.

Chapter 8

References

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- About Mumbai
- Google Earth Engine platform for classification
- Sentinel 2A data & Sentinel Bands
- Landsat 8 data & Landsat Bands
- Remote Sensing Indices
- The Elements of Statistical Learning Book by Jerome H. Friedman, Robert Tibshirani, and Trevor Hastie